

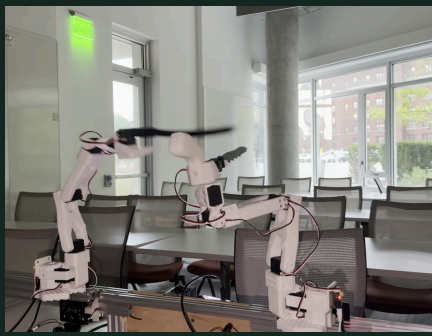
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Math and computer engineering at
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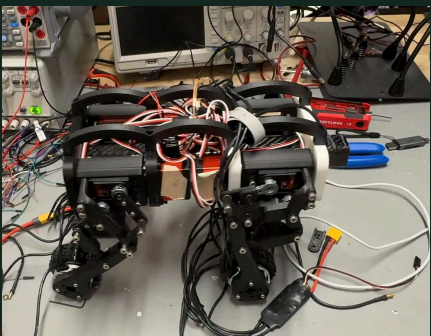
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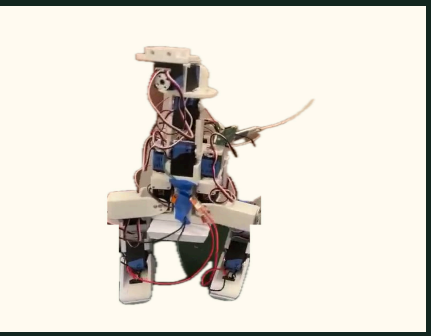
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01 **Robot Jousting** FALL 2026
Two real robot arms with 3D-printed swords fence each other on stepper rails while a crowd plays from their phones.



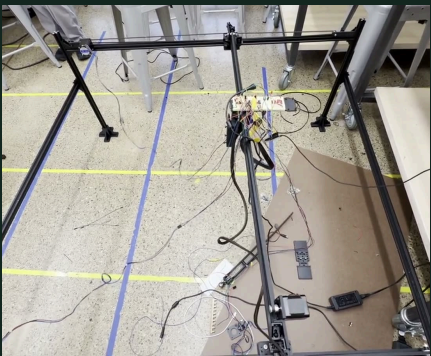
02 **Robot Dog** SUMMER 2026
A printed quadruped, from a single leg test to dancing on all fours.



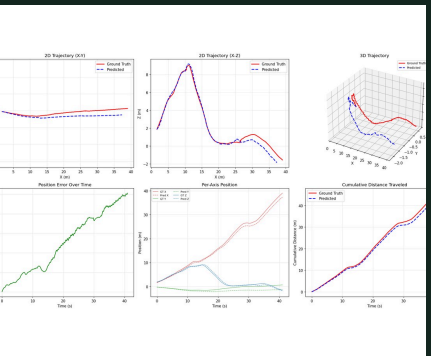
03 **Open Duck Mini** SUMMER 2025
An open-source biped built from bare servos up: soldered, wired, and driven until it walked and picked itself up off the floor.



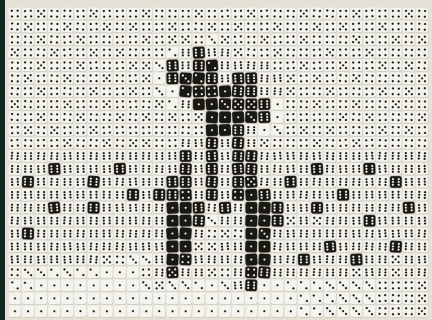
04 **Pi Day Pie Robot** SPRING 2026
A borrowed microwave, a printed arm, and a great deal of orange tape.



05 **Hack 100 Gantry** SPRING 2026
Built at Hack100 — partly self-designed CAD through to the first limit-switch test.



06 **TartanIMU** OCT 2025 – MAY 2026
Inertial odometry: a trajectory estimated from IMU alone, with a live demo on Hugging Face.



07 **Dice Arms** SUMMER 2025
A dice mosaic generated from the Duck render — each cell is one die, pip count tracking local tone.

chrisshi.com/robotics/

Portfolio · September 2026
Chris Shi · chrisshi@andrew.cmu.edu

The Tilt of Tiltford. Two real robot arms with 3D-printed swords fence each other on stepper rails while a crowd plays from their phones.

Two real SO-100/SO-101 robot arms, each holding a 3D-printed sword, fence each other on stepper rails at a medieval festival. Players fight from their phones. A crowd watches. It is an HRI experiment in whether a machine can make a room of people cheer. We built it at HackCMU 2026, on 12 September 2026.

Every move was designed in MuJoCo, then tuned on the real arms, and the moves are chained together by a transition library so one motion can run into the next. The game waits on the real motions — the screen does not resolve an exchange until the arms have actually fought it.

HACKCMU 2026

SO-100 AND SO-101 ARMS

STEPPER RAILS

MUJOCO

3D-PRINTED FENCING GRIPPER

TEAM OF THREE

CODE

github.com/avnithv/robot-jousting

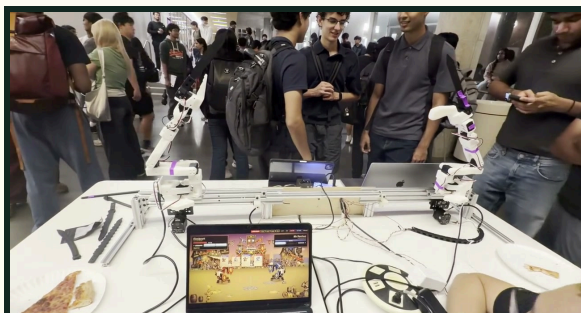
WRITE-UP

chrisshi.com/robo-jousting/

TRAILER

youtu.be/0l8j64tkMzg

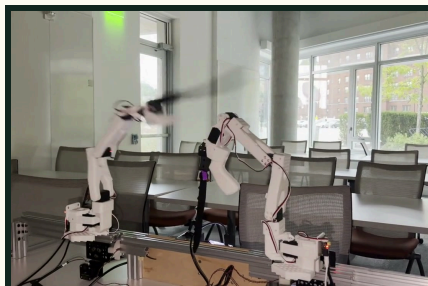
DEMO

youtu.be/c6h_7mXpHWY

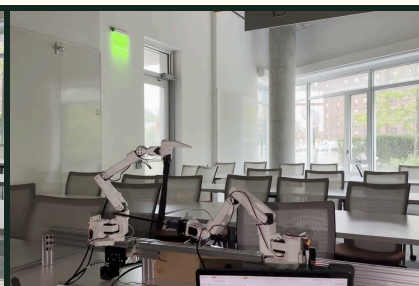
HackCMU 2026.

**Overhead chop, high bar**

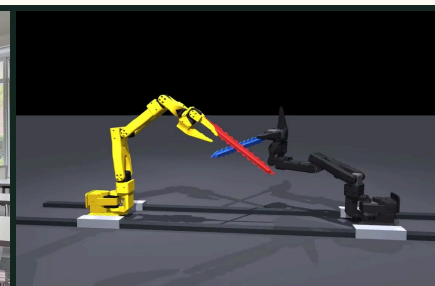
Overhead chop meets the high bar. Blocked.

**Overhead chop, low guard**

Overhead chop against a low guard. The hit lands.

**Low slash, right-hand guard**

A low slash from the left meets the right-hand guard. Blocked.

**Designed in simulation**

MuJoCo twin of the first exchange: the overhead chop into the high bar.

A printed quadruped, from a single leg test to dancing on all fours.

A printed quadruped. The structural parts are 3D printed, many of them self-designed. Each leg is driven by two servos mounted in the body, through pushrod linkages, from a PCA9685 servo controller on a Raspberry Pi. On the bench it stands and runs an authored dance routine on all four legs, powered over a tether.

CAD

No servo is mounted on a leg. Both actuators for each leg sit in the body and drive the lower link through a pushrod, so what swings is printed plastic and bearings. Moment of inertia scales with the square of distance from the joint, so a servo at the knee costs several times what the same mass costs at the hip. Hobby servos have a fixed torque budget, and leg inertia sets how fast a leg can move before it stops tracking its commands.

Prototyping

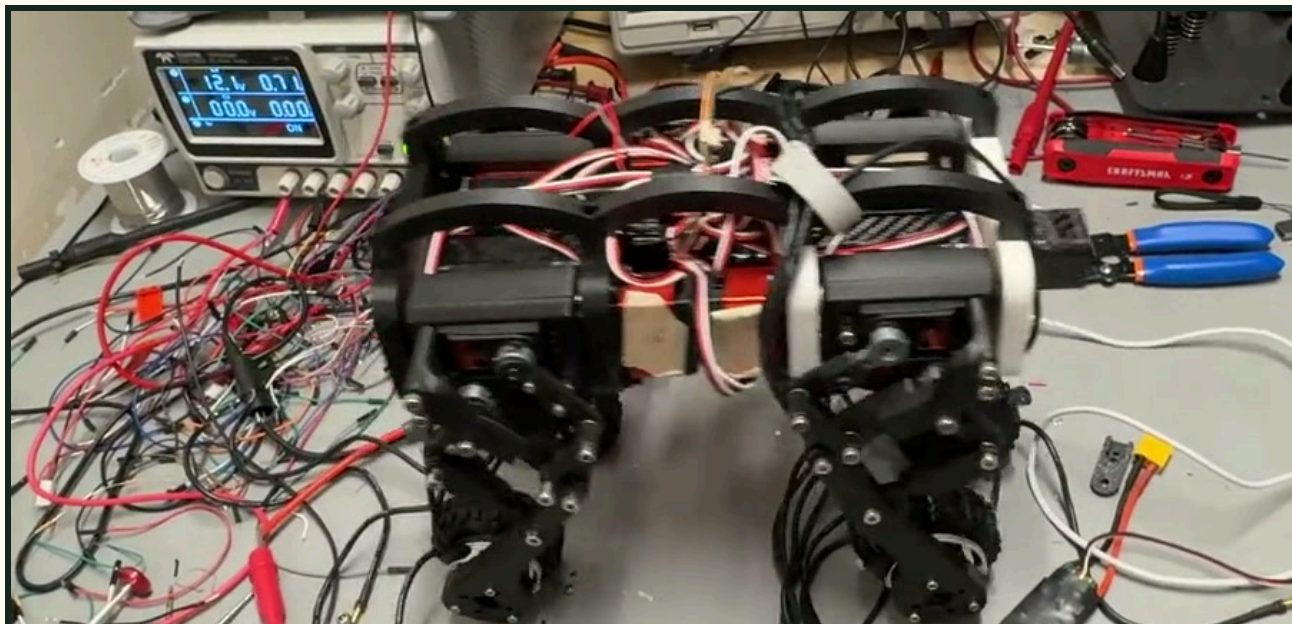
One leg was built and tested before the other three: a single leg bolted to a vertical extrusion on a plywood stand, driven from a breadboard with a dev board and a bench power module.

The rig answered three questions: whether the linkage moves through its full range without self-interference, whether the printed pin joints survive repeated load, and what the servo delivers against the real leg rather than the datasheet.

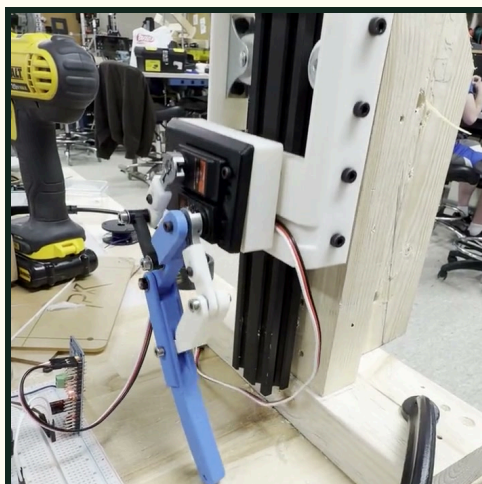
Robot animation

The dance is authored motion: a sequence of poses with interpolation between them, not a learned or planned gait.

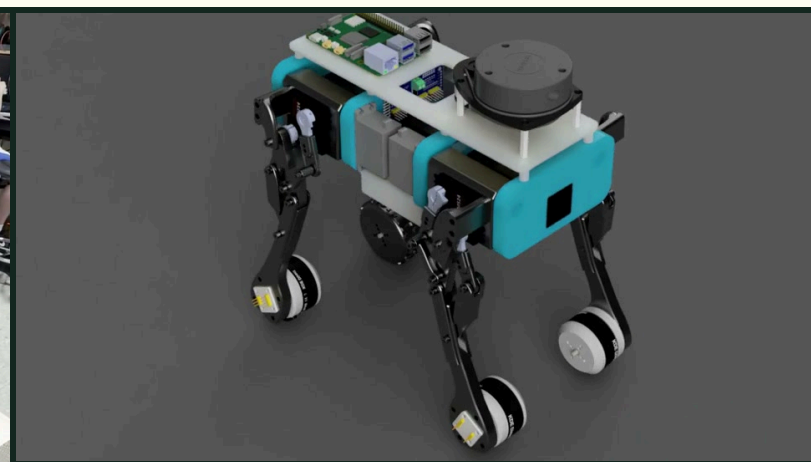
Every joint takes a position command and nothing else. No velocity, no torque, no feedback.



Final dancing



Leg test



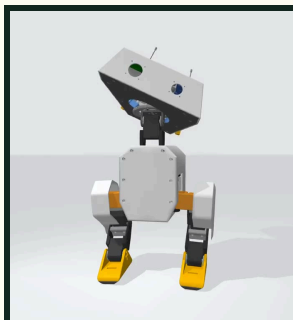
The CAD assembly. Two servos per leg sit inside the body, the Raspberry Pi and the RPLIDAR A1 ride on the top plate.

An open-source biped built from bare servos up: soldered, wired, and driven until it walked and picked itself up off the floor.

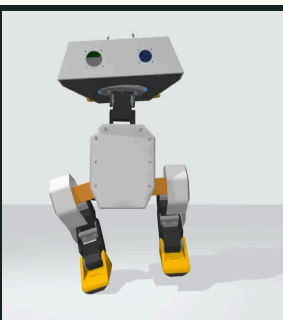
OPEN DUCK MINI

github.com/apirrone/Open_Duck_Mini

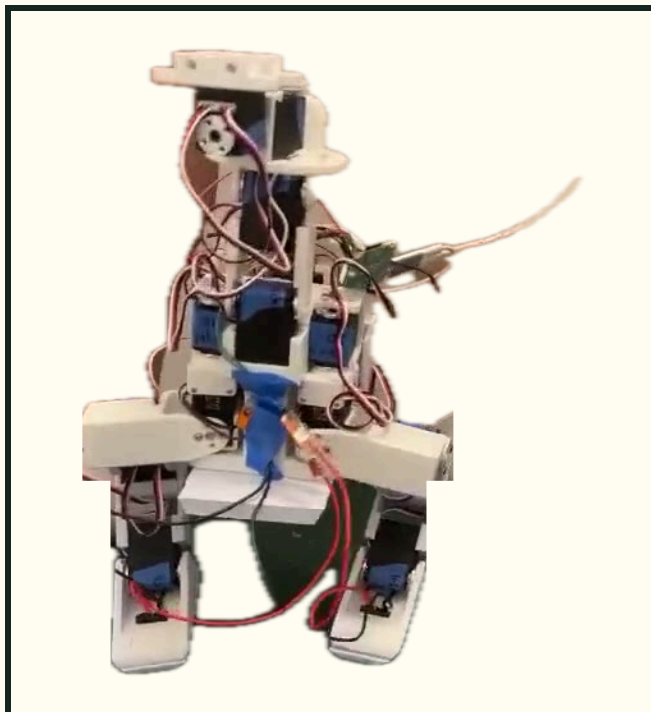
Duck renders as an HCI project



MuJoCo render



MuJoCo render

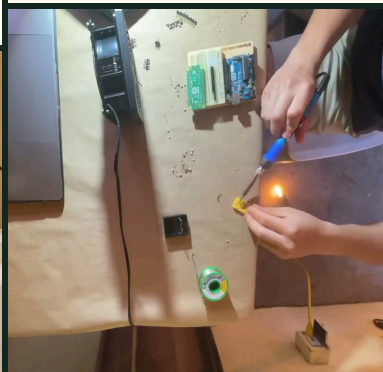


The build

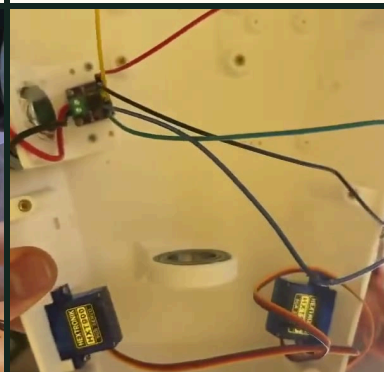
01 Disassembling Arms and Configuring Servos



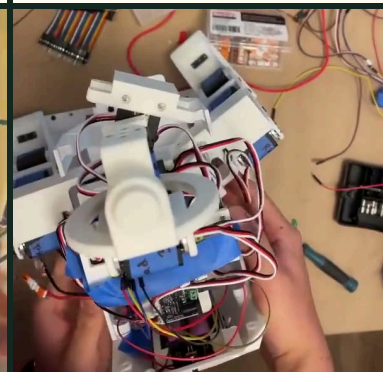
02 Soldering Pi0 2W



03 Soldering Microphone



04 Assembling Battery Pack



05 Test Runs

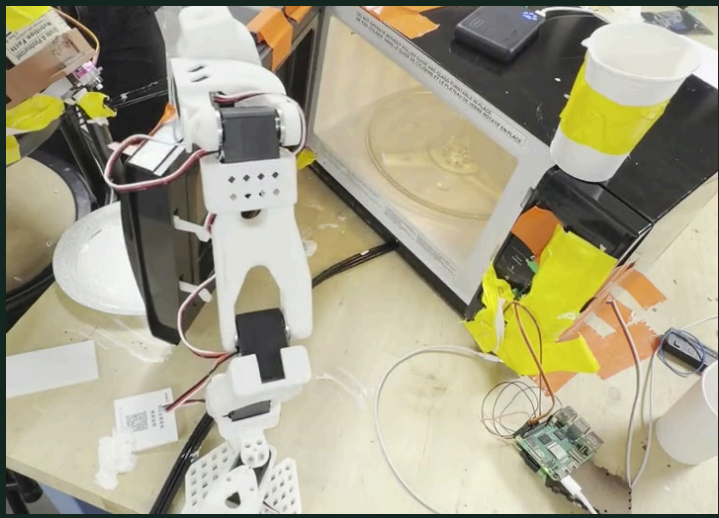


A borrowed microwave, a printed arm, and a great deal of orange tape.

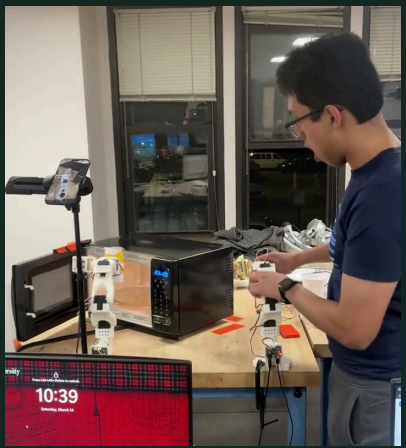
A Pi Day hackathon sprint. Each cup is a pie assembled ahead of time — a crumb base of crushed crackers under pumpkin pie filling — so it only needs heating. Everything after that is the robot's job: the door, the cup, the start button, and the whipped cream on top.

The microwave is stock hardware with no electrical interface, so every input had to be made mechanically. Servos driven by an ESP32 work the door and the start button, and every bracket holding them — along with the fixture that clamps the whipped cream can upside down above the plate — is self-designed and 3D printed.

- PI DAY HACKATHON
- ESP32
- SELF-DESIGNED 3D-PRINTED MOUNTS
- SERVO ACTUATION
- TWO-STAGE DOOR OPEN



Two-stage door open



Collecting training data: walking the arm through the motion by hand, repeatedly, so it can do it on its own.

01 Open TWO MOVES	02 Load CUP IN	03 Close DOOR SHUT	Retry THE TAPE GAVE UP	04 Start SERVO	05 Unload CUP OUT	06 Top WHIPPED CREAM
The gripper pulls the handle until the latch pops, then reaches up and shoves the acrylic bar taped to the top of the door. The arm is too short to swing the door itself, so the door was given a longer handle.	Cup off the top of the microwave and onto the turntable.	The arm rises to near-vertical, reaches the top of the open door and pushes it until the latch catches.	The same close, twenty-three seconds earlier. The door swings almost shut, rebounds, and the tape holding the acrylic bar peels off. Someone re-tapes it and the next attempt holds.	A servo on a printed bracket presses the panel.	Back out and onto the paper plate, which is parked under the nozzle for a reason.	The reason. The gripper tips the lever on an upside-down can and the cup fills from empty.

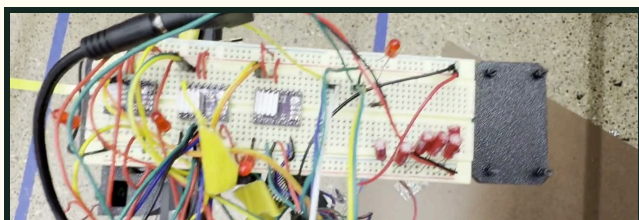
Built at Hack 100 — partly self-designed CAD through to the first limit-switch test.

A three-axis Cartesian gantry, about 1.2 m on a side, standing free on its own legs over a taped-out floor grid, built to reach any point in that grid, drop a tool head onto it, pick something up and put it somewhere else.

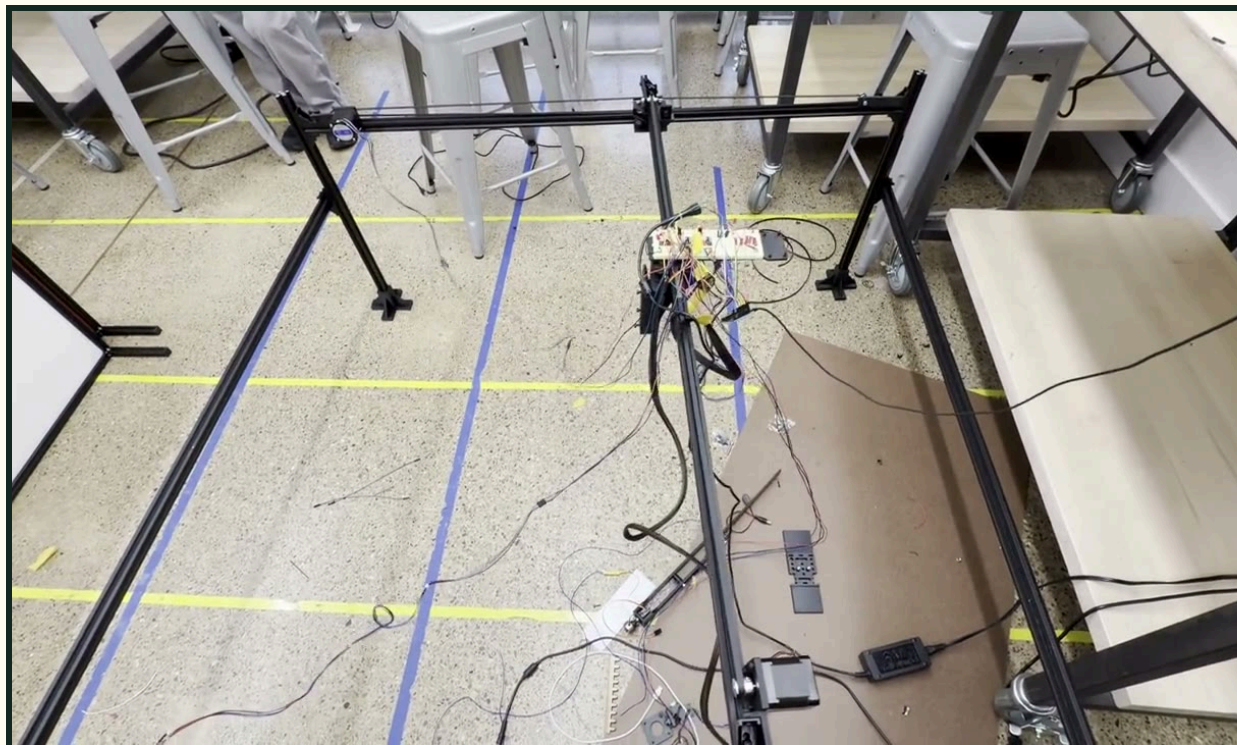
It was built in a week. 18-100 runs an optional hackathon in the week before spring finals — build anything you want — and rather than something that fits on a desk we went after a machine we could not comfortably finish. The single-axis bench test further down this page is dated 26 April and the assembled gantry running is 1 May. Everything here happened in the five days between them, off a 19-line bill of materials totalling \$352.50, with no kit and nothing bought pre-assembled. It won \$50, which did not cover the extrusion, and on the last evening we taught the steppers to play Ode to Joy.

Bringing it up

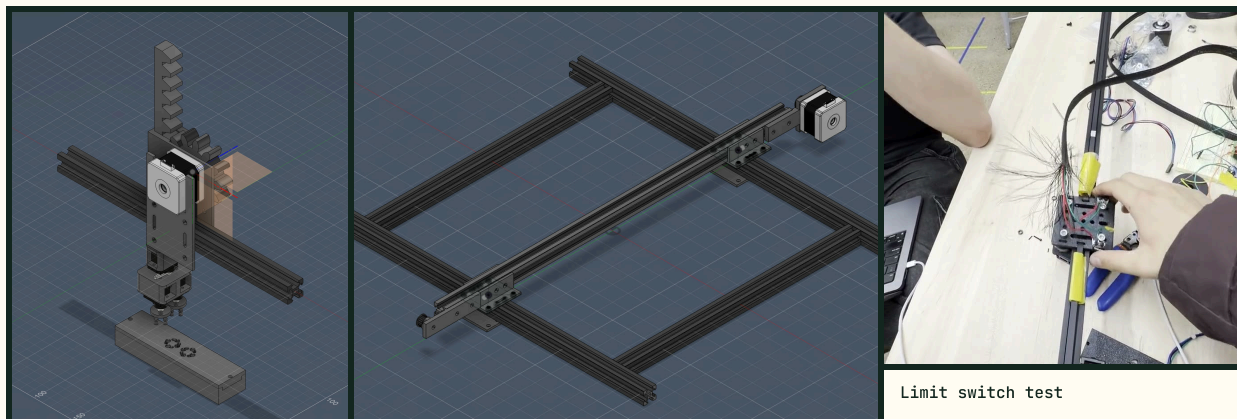
The limit switch clip is one rail on a bench — a single carriage, one motor, one switch — and it is dated five days before the machine ran. That order was the whole method, and on a week-long build it was the only affordable one. Every unknown in the machine — does the carriage bind, does the belt slip under acceleration, does the switch trigger repeatably at speed, does the driver stay cool — got paid for one at a time on a bench, so that assembling the full gantry was assembly and not discovery. There was no schedule in which to debug three axes at once for the first time.



Electronics



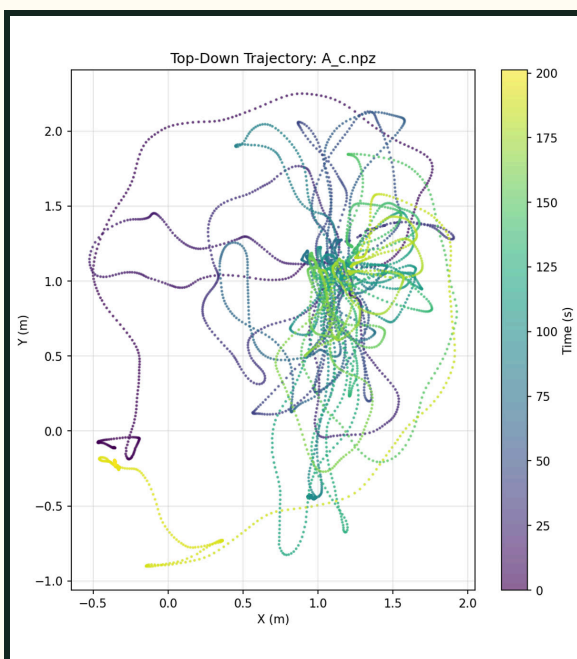
The finished machine, 1 May — the steppers are playing Ode to Joy.



Limit switch test

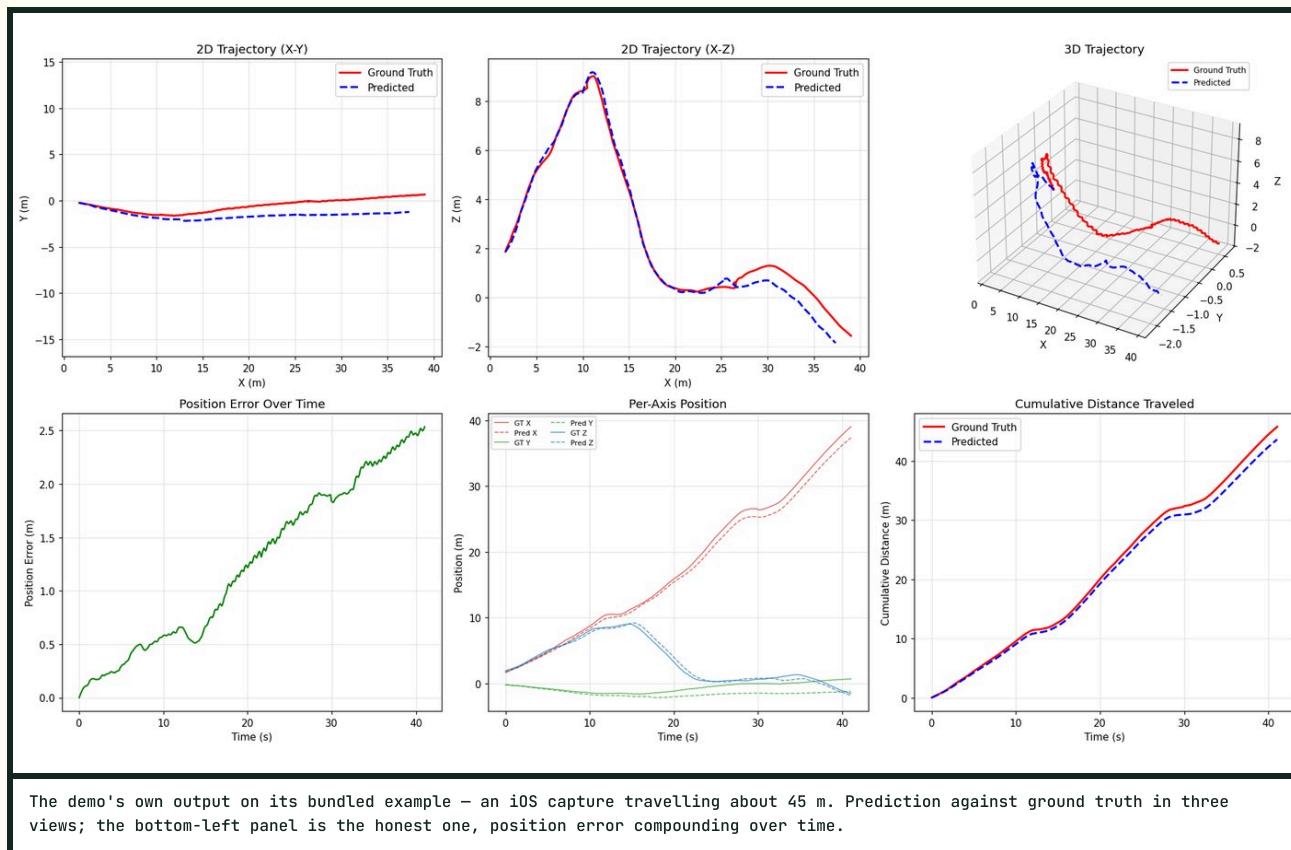
Inertial odometry: a trajectory estimated from IMU alone, with a live demo on Hugging Face.

TartanIMU is a learned inertial odometry project in the AirLab at Carnegie Mellon, where I did research from fall 2025 through the summer of 2026. The problem is odometry from an IMU alone: take the raw stream — three axes of acceleration, three of angular velocity — and recover the trajectory the sensor travelled, with no camera, no GPS, nothing else. The ambition is a foundation model for inertial data: one network trained across four motion types — car, dog, drone, human — rather than a model tuned per platform.



Looking at the data

The same sequence top-down, coloured by time.



The demo

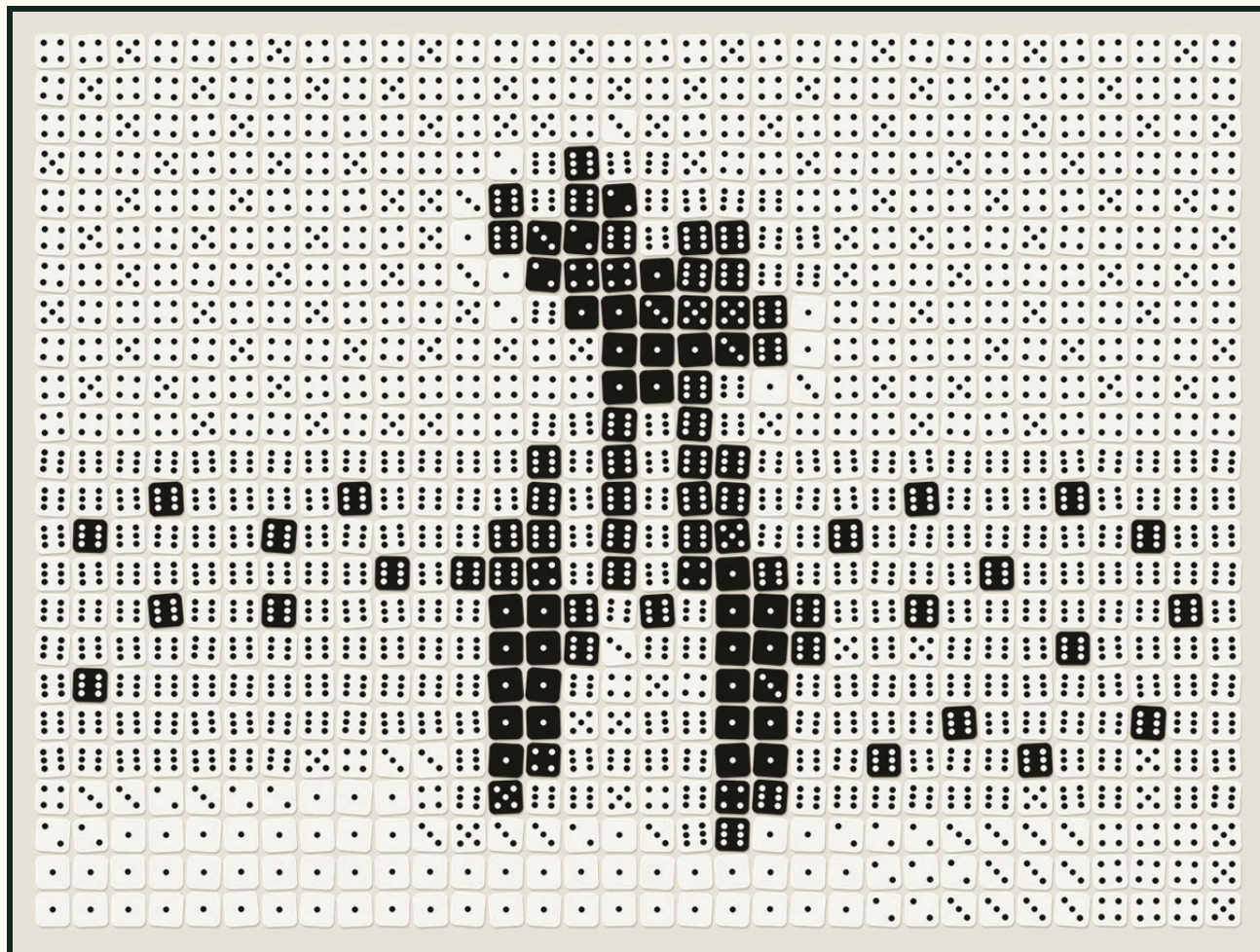
The public face is the Hugging Face Space linked on this page, implemented in early March: upload an npz in the project's format and it runs inference and draws what you see at the top of this page — the sensor streams, the predicted trajectory against ground truth, and the metrics. The model behind it is deliberately modest, an overfit LaMar model: a demonstration that the pipeline works end to end, not a claim about generalization.

HUGGINGFACE DEMO

The figures on this page are the demo's real output on its bundled example, an iOS capture from January 2022: ATE 1.4684 m, translational RTE 0.2044 m, drift RTE 0.2298 m, velocity RMSE 0.0903 m/s, inference in 0.35 s. The Space itself runs on the free CPU tier, which could not finish that example in six minutes, so these figures come from running the Space's own inference and plotting code on the 4090 — same code, same weights, same example, faster silicon.

huggingface.co/spaces/HongyiShi/TartanIMU-Demo

A dice mosaic generated from the Duck render — each cell is one die, pip count tracking local tone.



Under construction.